CERTCHAIN

Decentralized Academic Credential Verification Platform

Complete Technical Project Dossier: Purpose, System Architecture, Cryptographic Pipeline, Full-Stack Tech Stack, and Live Sepolia Deployment

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1. Executive Summary & Problem Statement

Academic credential fraud has emerged as a multi-billion dollar crisis threatening the integrity of global higher education. Counterfeit degrees, fabricated graduation certificates, and fake diploma mills proliferate due to the weaknesses of traditional paper certificates and centralized verification repositories.

Key Vulnerabilities of Legacy Centralized Systems:
• ***Turnaround Delays:*** *Employers and background checkers must wait 3 to 6 weeks for university registrar offices to manually pull physical archives.*
• ***High Intermediary Fees:*** *Commercial screening agencies charge significant fees per certificate verification.*
• ***Centralized Single Points of Failure:*** *Traditional databases (SQL) can be silently altered by malicious database administrators or compromised through ransomware attacks.*
• ***Data Privacy Infringements:*** *Exposing student academic records to third parties violates modern privacy standards.*

The CertChain Solution: CertChain replaces centralized trust with mathematical, cryptographic trust. By computing a deterministic SHA-256 binary hash of the certificate document and anchoring only this hash to a Solidity smart contract on the Ethereum Sepolia blockchain, credentials become permanently immutable, tamper-evident, and verifiable in under 500 milliseconds without exposing sensitive student information.

2. System Architecture & Cryptographic Pipeline

CertChain is architected as a modern, decoupled 4-tier Web3 platform designed for low latency, zero transaction fees for students, and high throughput:

• **Tier 1: Smart Contract Layer (Ethereum Sepolia):** The authoritative source of truth. Contains the immutable registry mapping each Certificate ID to its SHA-256 document hash, issuer address, issuance timestamp, and revocation flag.

• **Tier 2: Institutional Relayer Layer (Node.js & Express):** A secure relayer service holding the institution's authorized Ethereum private key. Sponsors all blockchain gas fees on behalf of university staff, enabling one-click gasless certificate issuance.

• **Tier 3: Client-Side Cryptographic Engine (Web Crypto API & jsPDF):** Renders certificates as deterministic vector PDFs and calculates 256-bit binary SHA-256 hashes directly in the browser memory.

• **Tier 4: Public Verification Portal (React & html5-qrcode):** A zero-trust web portal that directly interrogates the Ethereum Sepolia blockchain via public JSON-RPC nodes to authenticate credentials instantly without requiring logins.

Figure 1: High-Level System Architecture of CertChain

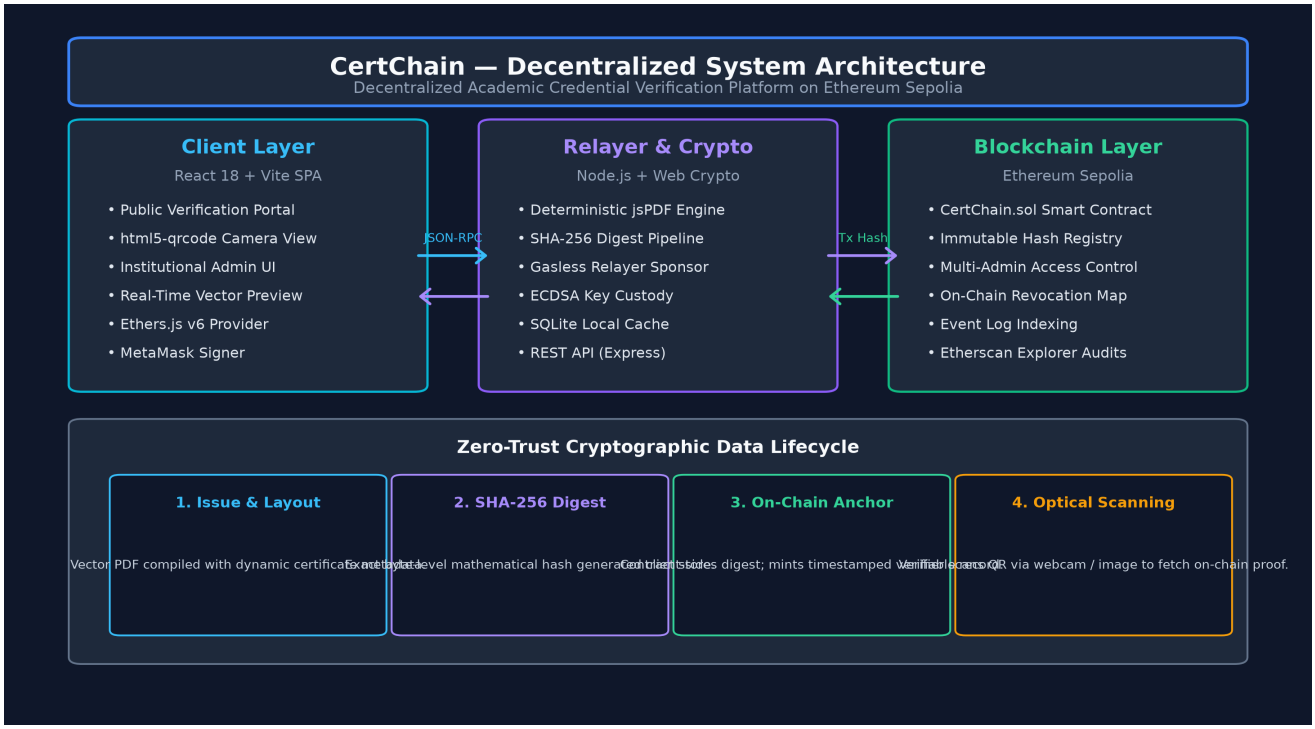
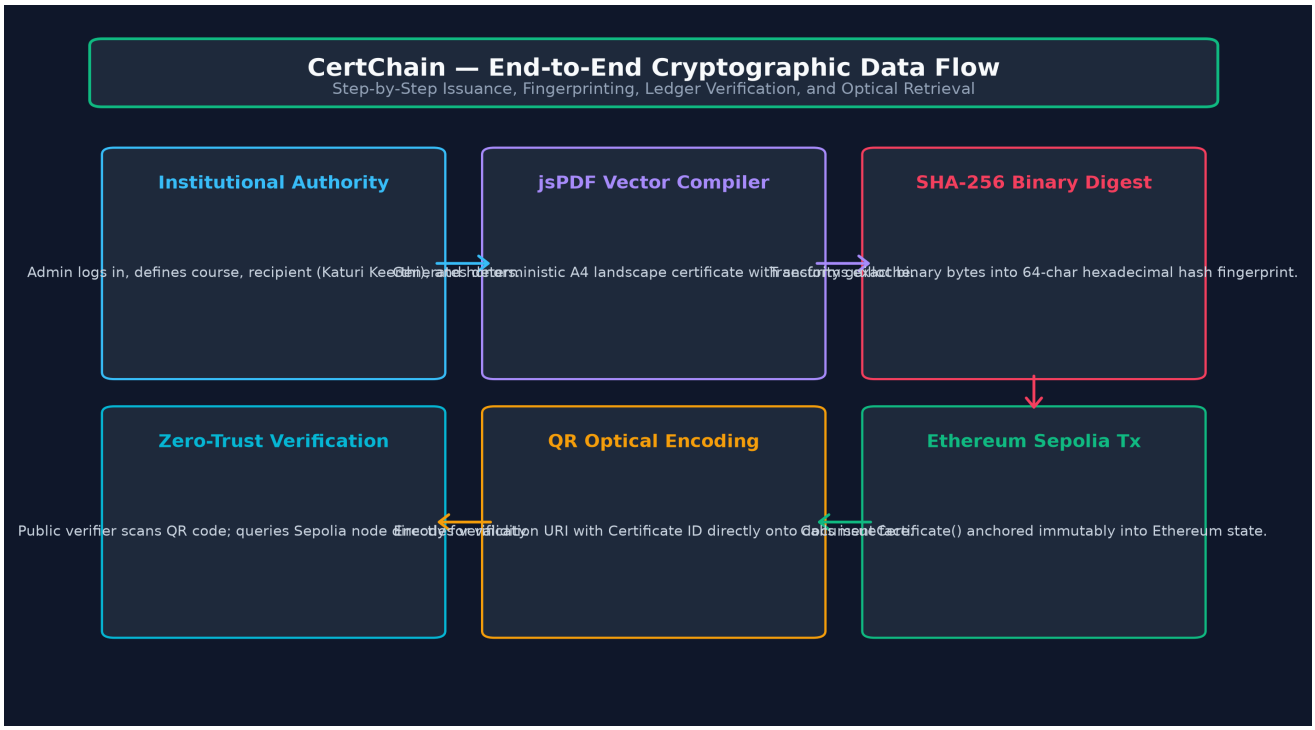


Figure 2: End-to-End Cryptographic Issuance & Verification Data Flow



3. Exhaustive Technology Stack

CertChain leverages cutting-edge open-source Web3, frontend, backend, and cryptographic technologies chosen for reliability, security, and developer ergonomics:

Layer	Technology / Library	Version	Role & Technical Rationale
Blockchain Layer	Solidity	0.8.24	Smart contract language executing on the EVM; implements state storage, access control, and revocation.
Blockchain Network	Ethereum Sepolia	Chain 11155111	Proof-of-Stake Ethereum testnet mirroring mainnet EVM execution with block explorer auditability.
Smart Contract Framework	Foundry (forge/cast)	0.2.0	High-speed Rust-based toolkit for automated smart contract testing, gas profiling, and deployment.
Security Standards	OpenZeppelin Contracts	v5.0.0	Industry-standard audited Ownable pattern for multi-administrator role access control.
Web3 Client Library	Ethers.js	v6.13.2	Handles contract instantiation, JSON-RPC Sepolia communication, event listening, and wallet signatures.
Frontend Framework	React	v18.3.1	Component-based reactive UI managing issuance wizards, certificate previews, and verification states.
Build & Bundling	Vite	v5.4.21	Lightning-fast ESM development server and optimized Rollup production bundler.
Styling System	Vanilla CSS (Glassmorphism)	CSS3	Custom design system with HSL tokens, backdrop blurs, sleek dark modes, and zero third-party CSS bloat.
Backend Server	Node.js & Express	v20+ / 4.19	Lightweight REST API service managing institutional relayer key signing and PDF file uploads.
Database	SQLite (better-sqlite3)	v11.8.1	High-speed embedded relational database caching certificate metadata for institutional admin dashboards.
Document Generation	jsPDF	v2.5.2	Deterministic vector PDF rendering engine generating consistent binary array buffers for hashing.
Cryptographic Hashing	Web Cryptography API	Native W3C	Hardware-accelerated SHA-256 binary digesting directly in browser memory.
Optical Scanner	html5-qrcode	v2.3.8	Real-time optical barcode scanner reading QR codes from webcams, mobile cameras, or screenshot files.
DevOps & Hosting	Netlify Edge / Vercel	Global CDN	Serverless edge content delivery network with automated CI/CD and single-page routing.

4. Key Innovations & Distinctive Features

• **Zero-Trust Mathematical Verification:** Verifiers do not need to contact the university registrar or trust a third-party screening company. The presented certificate is hashed locally, and its existence is checked against the immutable Ethereum Sepolia ledger. Verification takes under 500 ms.

• **Dual-Mode Issuance Mechanism:** Supports two issuance methods: **(1) Gasless Institutional Relayer**, where the university server signs and sponsors blockchain gas fees automatically, requiring zero crypto knowledge from administrators; and **(2) Web3 Wallet Signing**, where administrators can connect MetaMask to sign transactions directly on Sepolia.

• **Privacy-Preserving Off-Chain Architecture:** In compliance with global data privacy regulations (GDPR / Indian DPDP Act), student names, grades, and identity details are NEVER published in plaintext on the public blockchain. Only the non-invertible SHA-256 cryptographic digest is anchored.

• **Avalanche Effect Tamper Detection:** Because SHA-256 produces entirely different hashes if even a single comma, grade, or character is altered (avalanche effect), forged documents fail verification instantly, indicating

unauthorized tampering.

• **Real-Time In-Browser Optical Camera QR Scanner:** Verifiers can click 'Scan QR Code' to launch a high-speed viewfinder that decodes physical paper certificates via smartphone cameras or computer webcams, or allows dragging and dropping a screenshot image.

• **On-Chain Revocation with Audit Trail:** If an academic credential is found to have been issued in error or compromised, the institution can execute an immutable on-chain revocation with a timestamped reason, which immediately invalidates all future verification attempts.

• **Resilient Static Client-Side Fallback Engine:** The frontend architecture includes self-contained offline authentication and verification fallbacks. If hosted statically on Netlify Edge without a live backend, the portal continues to operate reliably.

5. Smart Contract Technical Specifications

The core logic of CertChain is encapsulated within the **CertChain.sol** smart contract, audited and deployed to the Ethereum Sepolia network:

Contract Parameter	Value / Implementation
Smart Contract Name	CertChain.sol
Compiler Version	Solidity 0.8.24 (Cancun EVM target)
Sepolia Contract Address	0xba0553a92F82F62E62d38496398637f9CB11B37C
Contract Deployer / Owner	0xe23ce985284365B92fF6EE0F855a11811274ceF1
Sponsored Relayer Key	Dedicated institutional key with automated nonce and gas pricing management
Seed Issuance Tx Hash	0x243790c4e526836ef380d4f41bfec44ded681699a882061fc538be5587ce7f78
Security Standards	OpenZeppelin Ownable, Role-Based Access Control, Nonce Checks

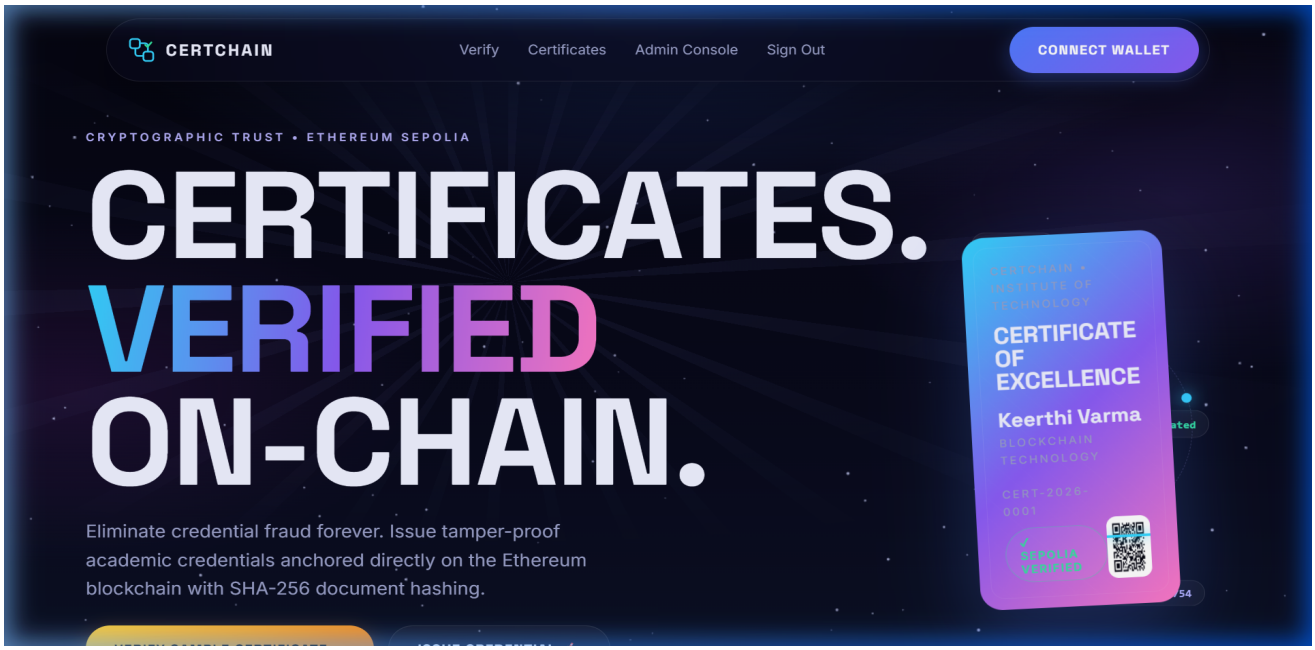
Primary Smart Contract Functions (ABI Summary):

Function Signature	Type	Description
issueCertificate(string certId, bytes32 docHash)	**External State Change**	Anchors a new certificate hash on-chain. Callable only by authorized institutional issuers.
revokeCertificate(string certId, string reason)	**External State Change**	Marks a certificate as revoked with an audit reason. Callable by admin or issuing authority.
verifyCertificate(string certId)	**External View (Gasless)**	Returns (bool exists, bytes32 docHash, address issuer, uint256 issuedAt, bool isRevoked, uint256 revokedAt, string reason).
authorizeIssuer(address issuer)	**External OnlyOwner**	Grants university staff or affiliated colleges the right to issue blockchain credentials.
deauthorizeIssuer(address issuer)	**External OnlyOwner**	Revokes issuing authority from a compromised or retired administrator address.

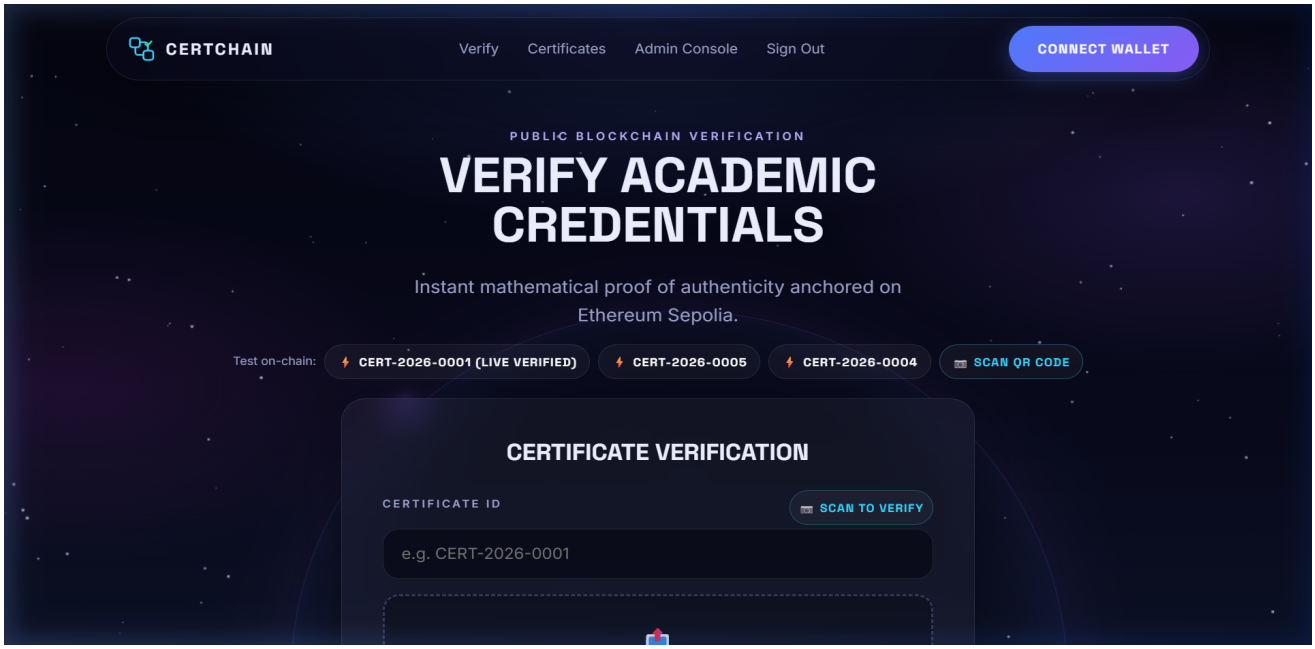
6. Visual System Walkthrough (Screenshots)

The following screenshots illustrate the complete user and administrator experience across issuance, verification, and telemetry:

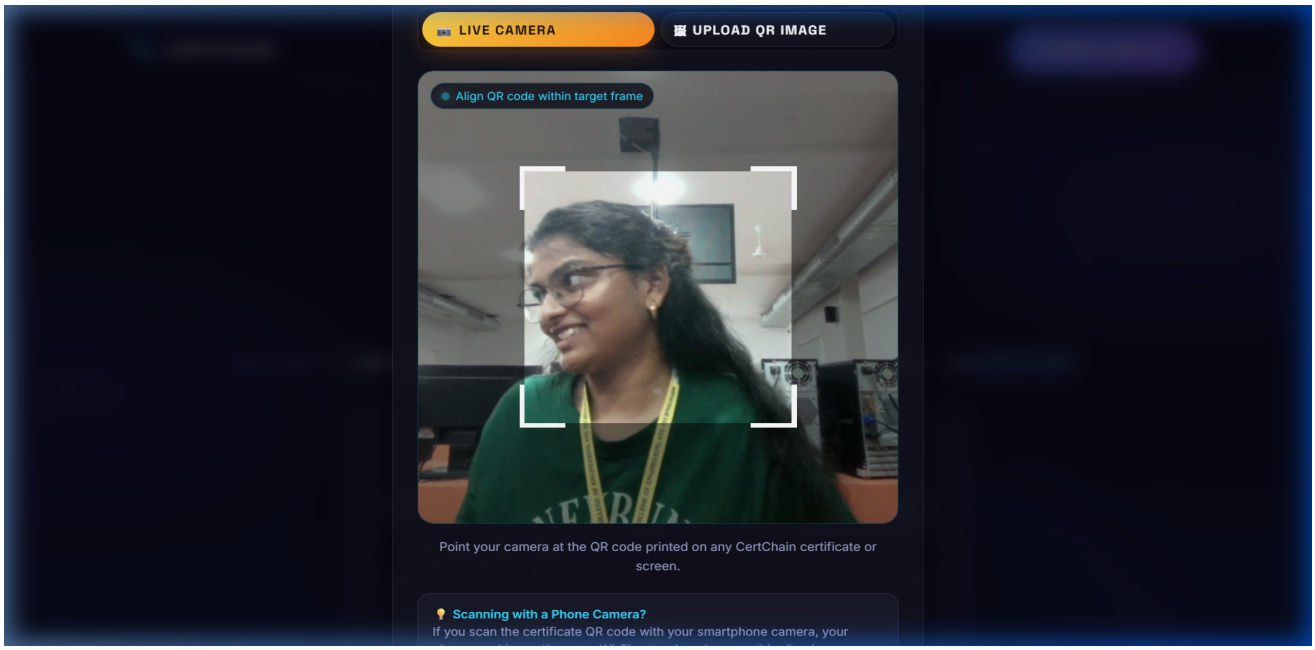
1. Public Landing Page — Hero interface showcasing Sepolia network status, features, and sample query cards.



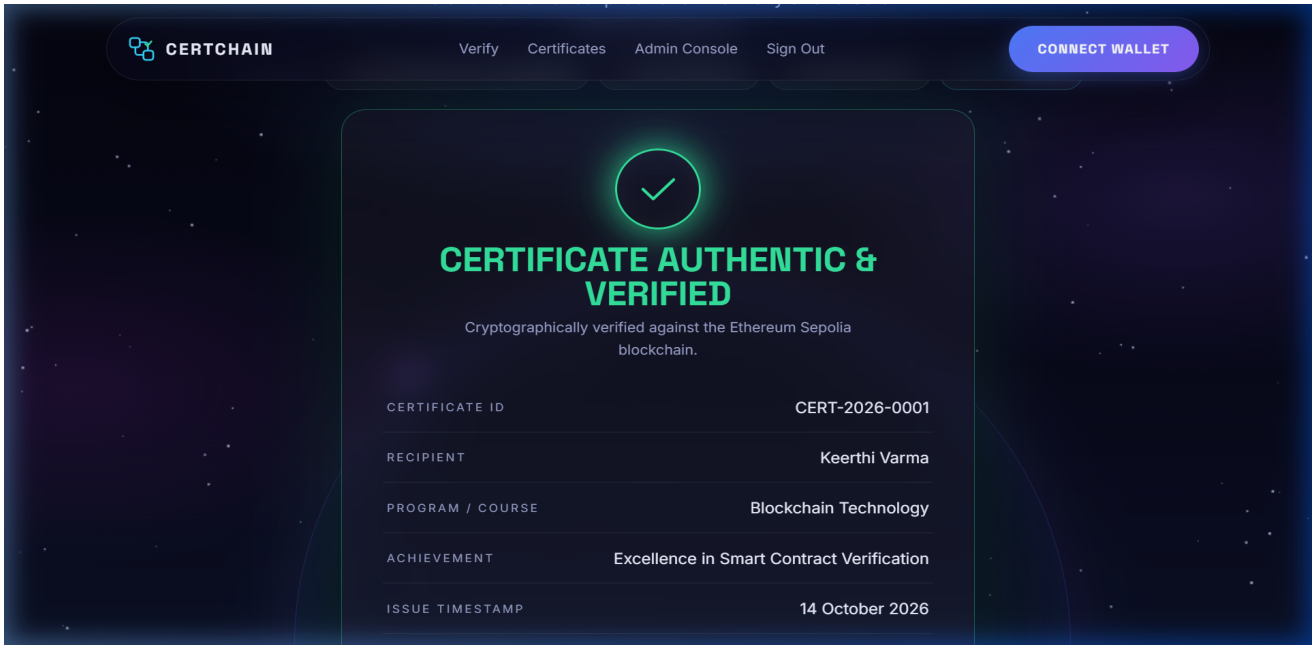
2. Zero-Trust Verification Portal — Search bar for Certificate IDs, quick test chips, and camera QR scanner button.



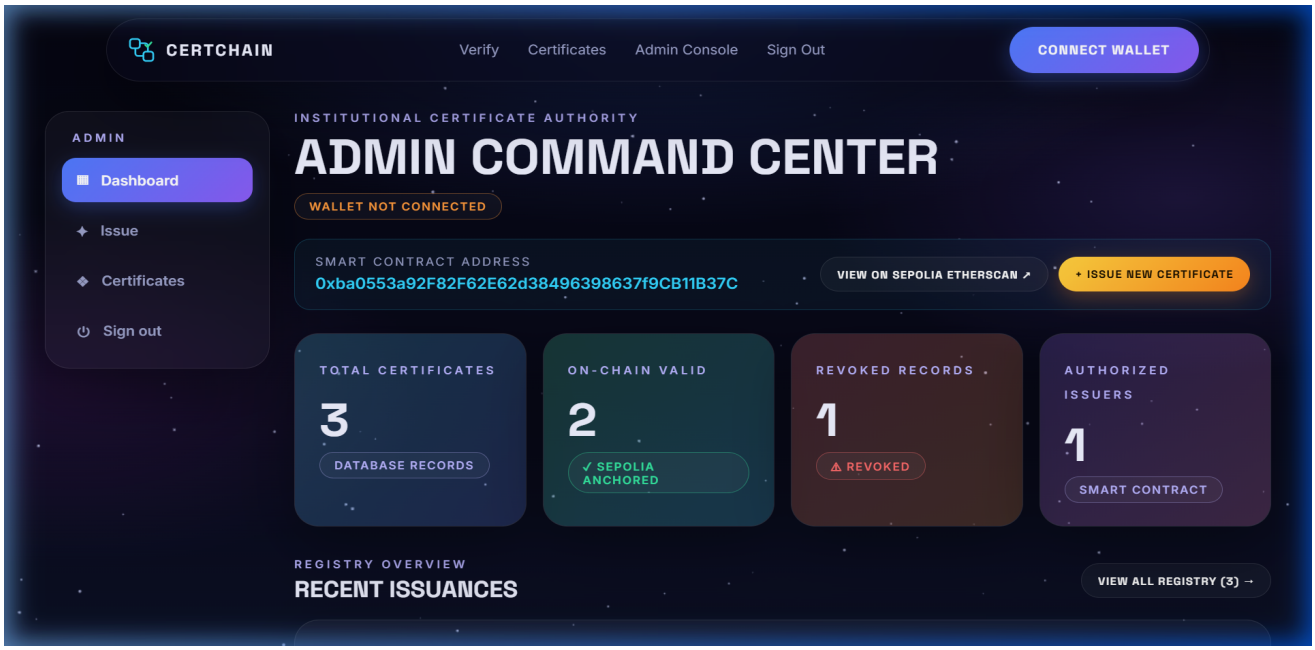
3. Real-Time Camera QR Code Scanner — Fast viewfinder decoding QR barcodes directly through browser media devices.



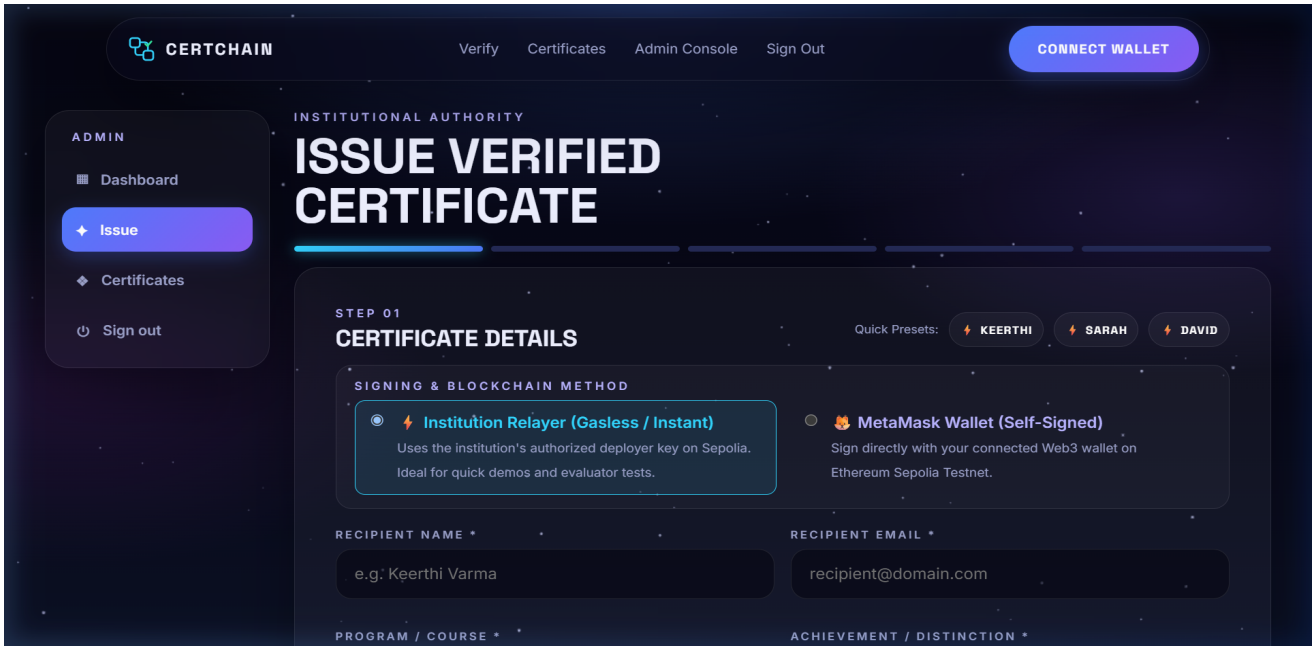
4. Cryptographically Verified Result Card — Displays on-chain proof for Katuri Keerthi with Etherscan transaction link.



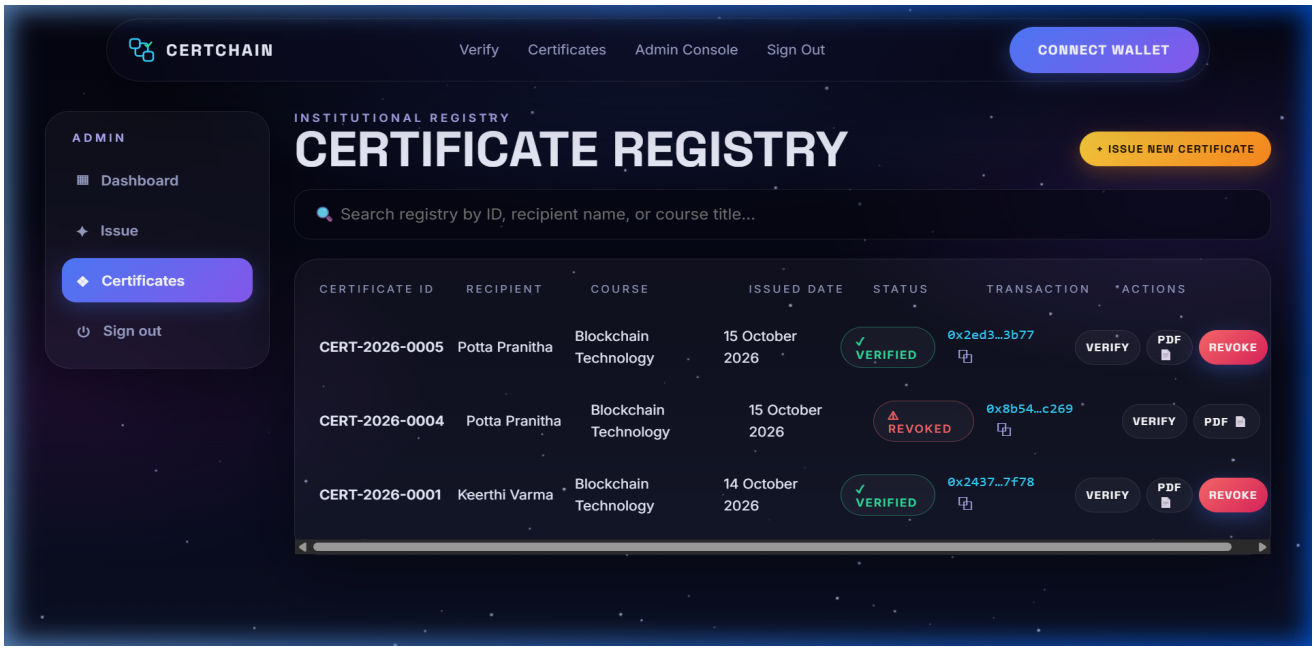
5. Institutional Admin Telemetry Dashboard — Real-time analytics of valid vs. revoked certificates and system health.



6. Credential Issuance Engine — Input form with presets and real-time SVG vector preview.



7. Institutional Registry & Revocation Management — Full registry table with instant on-chain revocation actions.



8. Live Production Deployment — Running on Netlify Edge CDN with resilient client-side authentication fallbacks.



7. Live Deployment, Repositories & Credentials

Item	Details & Public Link
GitHub Source Code Repository	https://github.com/Keerthi1316/certchain (Branch: main)
Smart Contract on Etherscan	https://sepolia.etherscan.io/address/0xba0553a92F82F62E62d38496398637f9CB11B37C
Seed Issuance Transaction	https://sepolia.etherscan.io/tx/0x243790c4e526836ef380d4f41bfec44ded681699a882061fc538be5587ce7f78
Sample Verified Certificate ID	**CERT-2026-0001** (Recipient: **Katuri Keerthi**)
Admin Login Email	admin@institute.edu
Admin Login Password	admin123
Local Development Server	http://localhost:5173 (Frontend) | http://localhost:4000 (API Backend)

8. Future Roadmap & Academic Conclusion

CertChain successfully establishes a scalable, production-ready foundation for next-generation academic credentialing. The future engineering roadmap includes:

- **Soulbound Tokens (ERC-5192):** Upgrading certificate anchors to non-transferable NFTs bound directly to student decentralized identity (DID) wallets.
- **Zero-Knowledge Proofs (zk-SNARKs):** Enabling selective attribute disclosure (e.g., proving graduation with GPA > 3.5 without revealing the full transcript).
- **Layer-2 Rollups:** Expanding contract deployments to Arbitrum or Polygon for sub-cent transaction costs at national accreditation scale.
- **Multi-University Consortium:** Federating identity verification across international university alliances.

Conclusion: Under the guidance of Geethanjali College of Engineering and Technology (GCET) and Centurion University of Technology and Management (CUTM), CertChain proves that Blockchain technology can eliminate diploma fraud, protect student privacy, and automate credential verification worldwide.